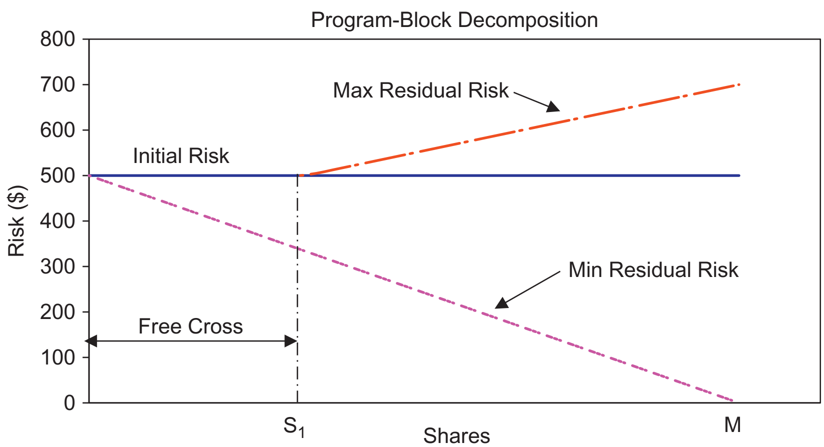
Notice in this case we make use of variance rather than standard deviation (square root) enabling the formulation of a quadratic optimization problem.

Figure 9.6 illustrates the program-block decomposition process. The graph shows the maximum and minimum residual risk that could arise for the corresponding number of shares that are entered into the dark pool or crossing network. The x-axis shows the number of shares from zero to M (the total number of shares). The graph shows three data series. The horizontal line is the initial risk of the portfolio. This is the residual risk that would arise if nothing is traded in the dark pool. The minimum residual risk line shows the best-case scenario for the corresponding number of shares. This is the residual risk if all shares are traded in the dark pool. This is a decreasing value. The maximum residual risk line shows the worse-case scenario that would arise if only some of the shares are entered into the dark pool. This is an increasing value. For example, suppose that an investor enters the entire hedged two-sided basket into a dark pool. If all shares are executed than the residual risk will be zero (since there are not any shares remaining). But if only one side of the portfolio is executed (such as all buy orders) the residual portfolio will have fewer shares but risk will be higher because the investor is no longer hedged to market movement.

In performing this exercise there will always be some number of shares S_1 such that the worse-case scenario will be equal to the initial risk value. This quantity S_1 is a "free block order" or "free crossing order" and represents the number of shares that can be entered into the dark pool



■ **Figure 9.6** Program Block Decomposition.